



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

CSE 4206: Digital Logic Design Lab
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Lab 5

Combinational Circuits II

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1 Introduction

In the previous lab session, we focused on the simulation of basic arithmetic circuits, specifically adders and subtractors. These circuits form the foundation of many digital systems. In today's lab, we will build upon that knowledge by simulating more complex combinational circuits. In this lab we will practice designing a few combinational circuits.

Warning: If you are caught copying another classmate's lab task, you will receive a zero for that specific lab.

2 BCD to 7 Segment Display

In today's lab, you will also need to display BCD (Binary-Coded Decimal) values on a 7-segment display. For this, you may use the 4511 BCD to 7-Segment IC.

The IC takes a 4-bit BCD input on pins A (pin 7), B (pin 1), C (pin 2), and D (pin 6):

- A is the Least Significant Bit (LSB)
- D is the Most Significant Bit (MSB)
- The outputs (pins 9 to 15) control the segments a to g of the 7-segment display.

Table 1: 4511 BCD to 7-Segment Display Driver – Key Pin Functions

Pin	Name	Set To	Description
1	B	Input	BCD Input Bit (Bit 1)
2	C	Input	BCD Input Bit (Bit 2)
3	LT	HIGH	Lamp Test – Keep HIGH to disable segment test mode
4	BI	HIGH	Blanking Input – Keep HIGH to enable output
5	LE	LOW	Latch Enable – Keep LOW for transparent input-to-output behavior
6	D	Input	BCD Input Bit (MSB – Bit 3)
7	A	Input	BCD Input Bit (LSB – Bit 0)
8	GND	GND	Ground connection (0V)
9–15	a–g	Output	Segment control outputs for 7-segment display
16	Vcc	+5V	Power supply

And for the seven segment display, make sure to ground (0) the 8th pin. Alternatively you can use something simpler like this:

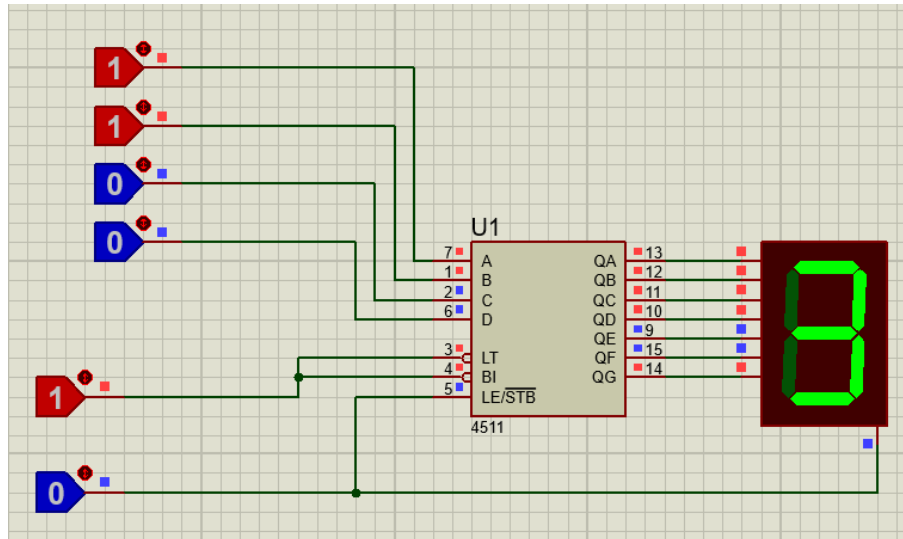


Figure 1: BCD to 7 segment display

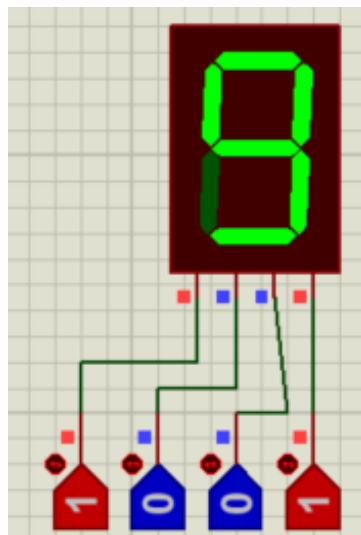


Figure 2: BCD to 7 segment display

2.1 Objectives

The objectives of this experiment are as follows:

- Perform 2-digit BCD addition with proper correction and display.
- Compare two BCD digits using basic logic gates.
- Use BCD to 7-segment decoder (e.g., 4511) to drive displays.

3 Tasks

Today's lab task are as follows -

3.1 Design a 4-bit BCD Adder Circuit

In this task, you will implement a **4-bit BCD adder circuit** and display both the **inputs and output using 7-segment displays**.

3.1.1 Instructions

1. Use two 4-bit BCD numbers (**A** and **B**) as inputs, where each input represents a valid decimal digit (0–9).
2. Construct a 4-bit BCD adder using 4-bit full adder blocks.
3. Implement the BCD correction logic so that when the binary sum exceeds 1001 (9) or produces a carry, the circuit adds 0110 (6) to obtain a valid BCD result.
4. Ensure that only valid BCD input combinations (0000–1001) are used for both **A** and **B** [you dont need to handle this].
5. Display both input values (**A** and **B**) and the output result on 7-segment displays.
6. Test the circuit with various valid input combinations to verify correct BCD addition operations.

Allowed Components

- 4-bit full adder
- Basic logic gates (AND, OR, NOT, XOR)
- BCD to 7-segment decoder modules
- Input switches or binary input sources
- 7-segment display units

3.2 BCD Comparator with 7-Segment Display Output

In this task, you will implement a **BCD Comparator** that compares three 4-bit BCD numbers and displays the largest value on a 7-segment display.

3.2.1 Instructions

1. Accept three 4-bit BCD inputs: $A[3:0]$, $B[3:0]$, and $C[3:0]$.
2. Compare the three BCD numbers and determine the largest value.
3. Display the largest number on a 7-segment display.
4. For example, if $A = 5$, $B = 7$, and $C = 2$, the 7-segment display should show 7.

Allowed Components

- Basic logic gates (AND, OR, NOT, XOR)
- Two-number comparator ICs
- BCD-to-7-segment decoder (or manual segment control using logic gates)
- 7-segment display
- Input switches or binary input sources

4 Instructions:

1. **File Naming Format:** ID_L05T01 and ID_L05T02.
2. For task 2 you have to come up with the explanation of how you are showing the result in a seven segment display.
3. You will get 1 hour to complete these tasks.
4. The rubrics for today's lab task can be found [here](#).